

Date

15/09/2023

Mathematics

Section A

1. 3) 3
2. 3) $\tan 60^\circ$
3. 2) 30°
4. 1) 10cm
5. 4) $a=0$
6. 4) Four decimal places
7. 3) 3 and 1
8. 2) $(0, -1)$
9. 4) No solution
10. 1) $k = -9/2$
11. 1) $k \neq -3$
12. 1) 8
13. 1) similar
14. 1) $x^3 y^4 z^4$
15. 3) $x^2/2 - x/2 - 6$
16. 3) intersecting or coincident
17. 1) Parallel
18. 4) 16:81
19. 3) Assertion (A) is true but Reason (R) is false
20. 1) Both A and R are true and R is correct explanation of A

Section B

$$21. \quad 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 \Rightarrow 6 \times 4 \times 3 \times 2 \times 1 \times 5 (1+1)$$

$$= 6 \times 4 \times 3 \times 2 \times 1 \times (2) \times 5$$

Since it has more than two factors it is composite.

22. $6x + 6y = 7 \Rightarrow 6x + 6y - 7 = 0$ ($a_1x + b_1y + c_1 = 0$)
 $kx + 18y = 21 \Rightarrow kx + 18y - 21 = 0$ ($a_2x + b_2y + c_2 = 0$)
 for infinite solutions $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
 So, $\frac{6}{k} = \frac{6}{18} = \frac{-7}{-21} \Rightarrow \frac{6}{k} = \frac{1}{3} \Rightarrow k = 12$

$\therefore$ Value of k is 12.

23. origin $(0,0)$ and point 2 is $(-6,8)$

By Distance formula $= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
 $\Rightarrow \sqrt{(0 - (-6))^2 + (0 - 8)^2}$
 $= \sqrt{36 + 64}$
 $= \sqrt{100} = 10$

So, distance is 10 units.

24. Given: $\sin 3A = \cos(A - 10^\circ)$

We know that $\sin \theta = \cos(90^\circ - \theta)$

So, $\sin 3A = \cos(90^\circ - 3A)$

$90^\circ - 3A = A - 10$ (from given)

$100^\circ = 4A \Rightarrow A = 25^\circ$ and $3A = 75^\circ$

So, $3A$ is an acute angle.

25. Given: $AB \parallel CD$

$\angle B = \angle C$ (Alternate Angle)

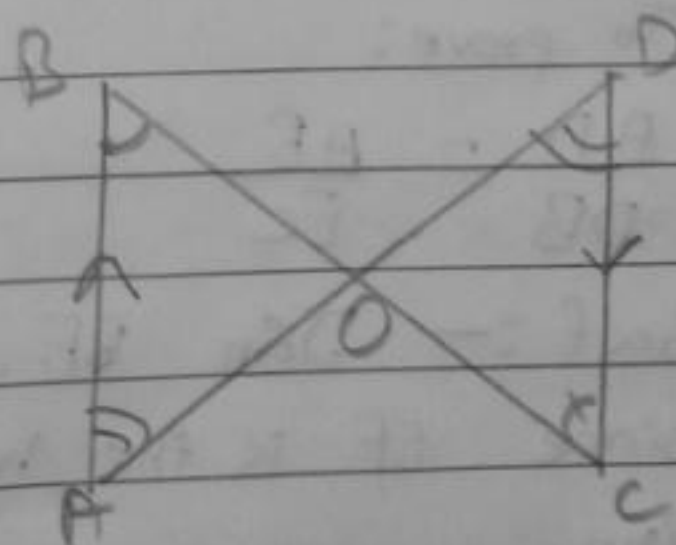
$\angle A = \angle D$ (Alternate Angle)

Proved above information for

$\triangle OAB$ and $\triangle ODC$

By AA rule

$\therefore \triangle AOB \sim \triangle DOC$



$\left. \begin{array}{l} \triangle OAB \text{ is same} \\ \text{as } \triangle AOB \text{ and } \triangle DOC \\ \text{is same as } \triangle DOC \end{array} \right\}$

Section C

$$\begin{aligned}
 26. \quad & \frac{\cos(90-\theta)}{1+\sin(90-\theta)} + \frac{1+\sin(90-\theta)}{\cos(90-\theta)} = 2 \operatorname{cosec} \theta \\
 = & \frac{\sin \theta}{1+\cos \theta} + \frac{1+\cos \theta}{\sin \theta} \quad \left\{ \begin{array}{l} \sin(90-\theta) = \cos \theta \\ \cos(90-\theta) = \sin \theta \end{array} \right\} \\
 = & \frac{\sin^2 \theta + (1+\cos \theta)^2}{\sin \theta (1+\cos \theta)} \\
 = & \frac{1-\cos^2 \theta + 1+\cos^2 \theta + 2\cos \theta}{\sin \theta (1+\cos \theta)} \quad \left\{ \begin{array}{l} \sin^2 \theta = 1-\cos^2 \theta \\ (\sin^2 \theta + \cos^2 \theta = 1) \end{array} \right\} \\
 = & \frac{2+2\cos \theta}{\sin \theta (1+\cos \theta)} \Rightarrow \frac{2(1+\cos \theta)}{\sin \theta (1+\cos \theta)} = \frac{2}{\sin \theta} \\
 = & 2 \operatorname{cosec} \theta \quad \left\{ 1/\sin \theta = \operatorname{cosec} \theta \right\}
 \end{aligned}$$

Hence Proved

27. Base Proportionality Theorem states that, if a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides in proportion for that triangle.

Let $\triangle ABC$, where line l is parallel to BC , intersect AB at D and AC at E .

To prove:

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Proof :- Join BE, CD and Draw $EF \perp AB, DG \perp CA$

since EF is the height of $\triangle ADE$ and $\triangle DBE$

Now,

$$\text{Ar}(\triangle ADE) = \frac{1}{2} (\text{Base} \times \text{Height}) = \frac{1}{2} \times AD \times EF$$

$$\text{Ar}(\triangle DBE) = \frac{1}{2} (\text{Base} \times \text{Height}) = \frac{1}{2} \times DB \times EF$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DBE)} = \frac{\frac{1}{2} \times AD \times EF}{\frac{1}{2} \times DB \times EF} = \frac{AD}{DB} \quad \dots (i)$$

Also,

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DCE)} = \frac{\frac{1}{2} \times AE \times DG}{\frac{1}{2} \times EC \times DG} = \frac{AE}{EC} \quad \dots (ii)$$

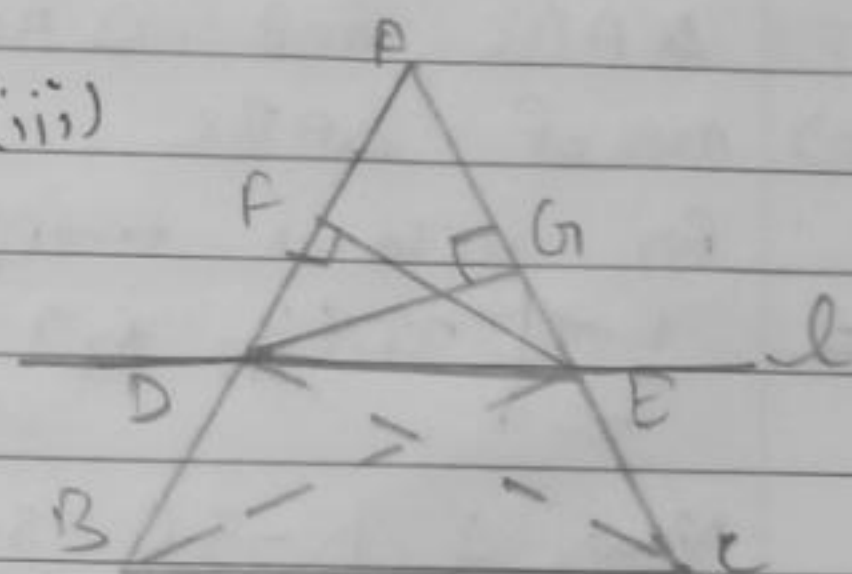
But $\triangle DBE$ and $\triangle DCE$ are on the same base DE and between the same parallels

$$\therefore \text{ar}(\triangle DBE) = \text{ar}(\triangle DCE) \quad \dots (iii)$$

From (i), (ii) and (iii) we have

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Hence Proved



28. let Sagar age be x and Tiru age be y
Given

$$x - 5 = 2(y - 5) \quad \dots (i)$$

$$x + 10 - (y + 10) = 10 \quad \dots (ii)$$

$$\text{So, } x - 5 - 2y + 10 = 10 \Rightarrow x - 2y = -5$$

$$x + 10 - y - 10 = 10 \Rightarrow x - y = 10$$

Now, By elimination

$$x - 2y = -5$$

$$- \underline{x - y = 10}$$

$$+y = -15 \Rightarrow y = 15$$

$$\text{Now, } x - 2(15) = -5 \Rightarrow x - 30 = -5 \Rightarrow x = 25$$

So, age of Sagar is 25 years

age of Tiru is 15 years

When Tiru have borned Sagar's age was 10 years.

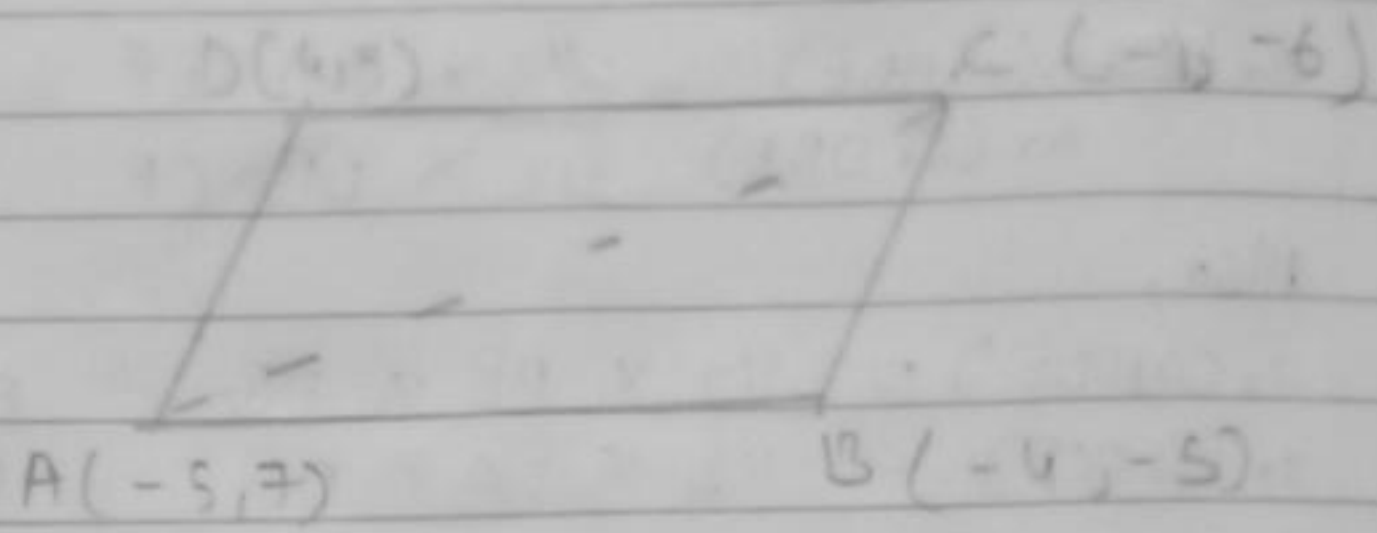
29. Given

A (-5, 7)

B (-4, -5)

C (-1, -6)

D (4, 5)



To find its area we will split it into 2 triangles $\triangle ABC$ and $\triangle ACD$

$\Rightarrow$ area of $\triangle ABC$

for area of triangle we have: -

$$= \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) |$$

$$\text{Now, } \frac{1}{2} | -5(-5 - (-6)) + (-4)(-6 - 7) + (-1)(7 - (-5)) |$$

$$= \frac{1}{2} | -5(1) + (-4)(-13) + (-1)(12) |$$

$$= \frac{1}{2} | -5 + 52 - 12 | = \frac{1}{2} | 35 | = 17.5 \text{ units square}$$

Now for $\triangle ACD$

$$\frac{1}{2} | -5(-6 - 5) + (-1)(5 - 7) + 4(7 - (-6)) |$$

$$\frac{1}{2} | -5(-11) + (-1)(-2) + 4(13) |$$

$$\frac{1}{2} | 55 + 2 + 52 | = \frac{1}{2} | 109 | = 54.5 \text{ units square}$$

So, Total area is 72 sq units.

30. HCF of 861 and 1353: -

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

So, HCF of 861 and 1353 is 123.

31. A point of x-axis means y-co-ordinate is 0
So it is in form of (x, 0)

Given

Distance of (x, 0) from (5, -2) and (-3, 2) is equal

By distance formula

$$\Rightarrow \sqrt{(5-x)^2 + (-2-0)^2} = \sqrt{(-3-x)^2 + (2-0)^2}$$

$$\Rightarrow \sqrt{(5-x)^2 + 4} = \sqrt{(-3-x)^2 + 4}$$

$$\Rightarrow (5-x)^2 + 4 = (-3-x)^2 + 4$$

$$\Rightarrow (5-x)^2 = (-3-x)^2$$

$$\Rightarrow \cancel{5-x} = \cancel{-3-x} \Rightarrow 25 + x^2 - 10x = 9 + x^2 + 6x$$

$$\Rightarrow 16 = 16x \Rightarrow x = 1$$

So, value of x is 1

Section D

32. $\tan A + \sin A = m$

$$\tan A - \sin A = n$$

$$m^2 = (\tan A + \sin A)^2 = \tan^2 A + \sin^2 A + 2 \sin A \tan A$$

$$n^2 = (\tan A - \sin A)^2 = \tan^2 A + \sin^2 A - 2 \sin A \tan A$$

$$m^2 - n^2 = (\tan A + \sin A)^2 - (\tan A - \sin A)^2$$

$$= 4 \sin A \tan A \quad \dots (i)$$

$$\text{Now, } mn = (\tan A + \sin A)(\tan A - \sin A)$$

$$= \tan^2 A - \sin^2 A$$

$$= \sin^2 A / \cos^2 A - \sin^2 A = \frac{\sin^2 A - \sin^2 \cos^2 A}{\cos^2 A}$$

$$\frac{\sin^2 A (1 - \cos^2 A)}{\cos^2 A} \Rightarrow \frac{\sin^2 A (\sin^2 A)}{\cos^2 A} \quad \left\{ 1 - \cos^2 \theta = \sin^2 \theta \right\}$$

P.T.O

$$\frac{\sin^4 A}{\cos^2 A} = mn \quad \text{and root } mn = \frac{\sin^2 A}{\cos A} \dots (ii)$$

from (i)

$$\sin A \cdot \tan A = \sin A \times \sin A / \cos A = \sin^2 A / \cos A \dots (iii)$$

So, LHS

$$= 4 \sin A \cdot \tan A \Rightarrow 4 \sqrt{mn} \quad \text{from (iii)}$$

Hence Proved

33. Given

$$x - y + 1 = 0 \quad \dots (i)$$

$$3x + 2y - 12 = 0 \quad \dots (ii)$$

Now,

$$x - y = -1 \quad \text{and} \quad 3x + 2y = 12$$

for the (i) equation.

$$x = -1 + y \Rightarrow x = y - 1$$

x	-1	0	1	2	3
y	0	1	2	3	4

for the (ii) equation

$$3x + 2y = 12 \Rightarrow 3x = 12 - 2y \Rightarrow x = 4 - \frac{2}{3}y$$

x	4	10/3	8/3	2
y	0	1	2	3

x, y intersects at 2, 3

Points on x-axis are: (-1, 0) and (4, 0)

So the area of triangle formed by it is: ~

$$\frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) |$$

$$\frac{1}{2} | -1(3 - 0) + 2(0 - 0) + 4(0 - 3) |$$

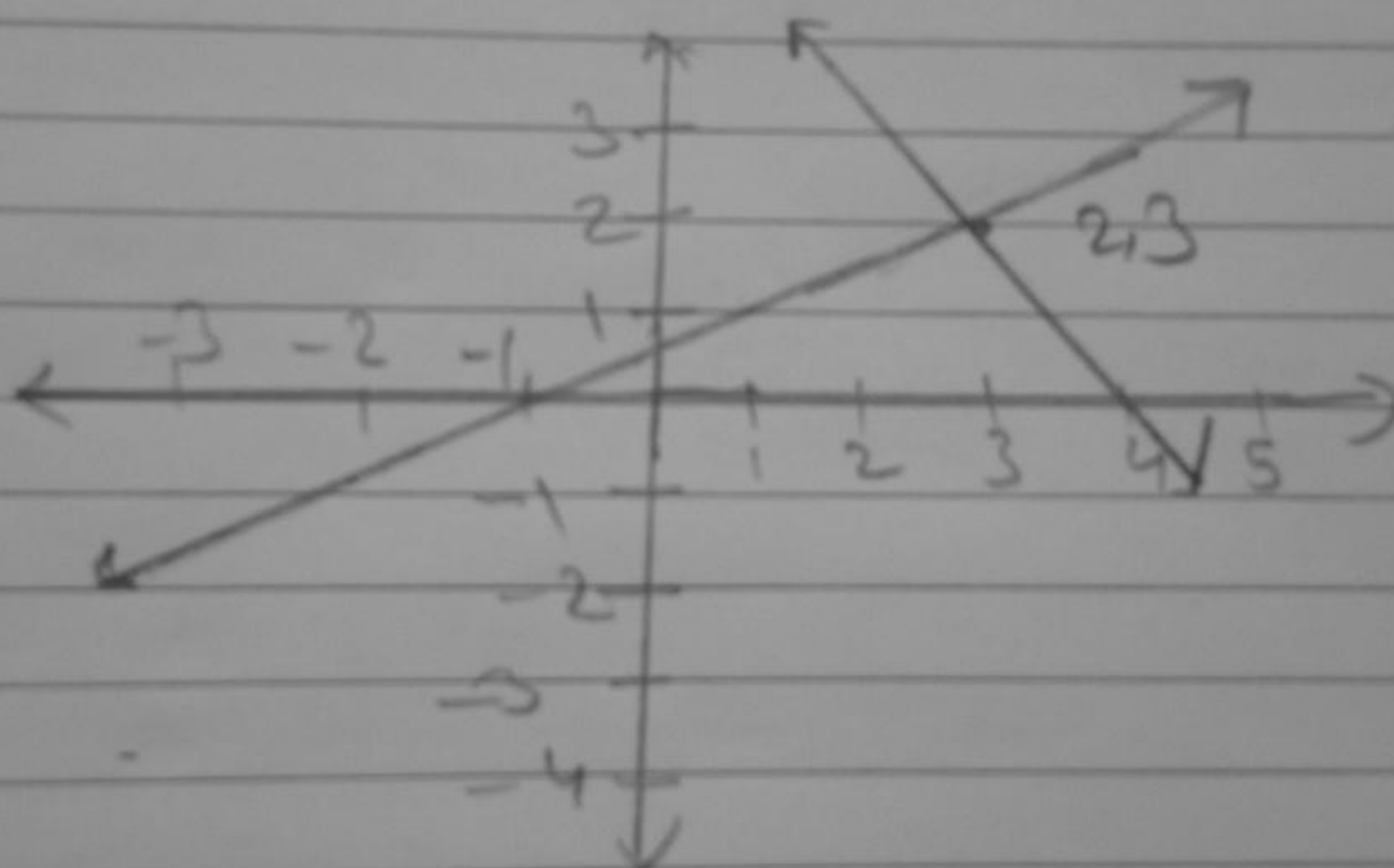
$$\frac{1}{2} | (-3) + 0 + (-12) | = \frac{1}{2} | -15 | = \frac{1}{2} (15)$$

Area of triangle formed by intersection and x-axis is 7.5 sq units.

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34. Given:

$$\text{Polynomial} = (2x^3 - 4x - x^2 + 2)$$

Two zeros are $\sqrt{2}$ and $-\sqrt{2}$

So,

we can write

$$x - \sqrt{2} = 0 \text{ and } x + \sqrt{2} = 0 \text{ as they are zero/roots}$$

$$\text{Now, } (x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$$

Now,

$$\begin{array}{r} x^2 - 2 \overline{) 2x^3 - 4x - x^2 + 2} \quad (2x - 1 \\ - \underline{2x^3 - 4x} \quad \downarrow \\ -x^2 + 2 \\ - \underline{-x^2 + 2} \\ 0 \end{array}$$

So, ~~$2x - 1$ is also a zero~~

$$(2x - 1)(x - \sqrt{2})(x + \sqrt{2}) = 2x^3 - 4x - x^2 + 2$$

$$\text{Now, } 2x - 1 = 0 \Rightarrow 1 = 2x \Rightarrow 1/2 = x$$

Therefore its all zeros are

$$1/2, \sqrt{2}, -\sqrt{2}$$

Since it has degree of 3 it have 3 zeros.

35. Given:

$$\text{Point A} = (4, -3)$$

$$\text{Point B} = (8, 5)$$

Ratio is which they divide 3:1

We need to find point C (Point which divides the line segment in 3:1)

So,

By Section formula

$$\left[\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \right), \left(\frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) \right] = C(x, y)$$

$$m_1 = 3, m_2 = 1, x_1 = 4, x_2 = 8, y_1 = -3, y_2 = 5$$

Now,

$$\left[\frac{3(8) + (1)(4)}{3+1}, \frac{3(5) + 1(-3)}{3+1} \right] = \text{Point } C(x, y)$$

$$= \left[\frac{24+4}{4}, \frac{15-3}{4} \right] = C(x, y)$$

$$= [7, 3] = C(x, y)$$

So the co-ordinates for C are (7, 3)

Section E

36.

(i) 2) (0, 12)

(ii) 1) (3, 5)

(iii) 3) 5m

37.

(i) 3) Parabola

(ii) 3) 0

(iii) 3) $-\frac{3}{2}$

38.

(i) 3) 24cm^2

(ii) 1) RHS

(iii) 4) Converse of Pythagoras Theorem